

Hyun-Jic Oh

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RESEARCH SUMMARY

AI/ML researcher specializing in diffusion-based generative models for medical image analysis and data-efficient visual learning. My work develops image synthesis, translation, and augmentation methods to build robust medical AI systems under limited data, class imbalance, weak annotation, and modality gaps, with growing focus on multimodal medical AI and intelligent systems.

EXPERIENCE

- Visual Computing Group, Harvard University** Mar. 2024 – Mar. 2025
Visiting Scholar Cambridge, MA, USA
 - Developed marker-wise conditioned diffusion models for virtual multiplex staining, translating routine H&E histology into high-plex immunofluorescence images for spatially resolved tissue analysis. (AAAI 2026)
 - Collaborated with researchers at Harvard SEAS, HMS/MGH, and systems biology groups on computational pathology, multimodal biomedical imaging, and translational medical AI projects.
- High Performance Visual Computing Lab, Korea University** Aug. 2020 – Aug. 2026
Ph.D. in Computer Science Seoul, South Korea
 - Developed diffusion-based data synthesis and image-label co-generation pipelines for nuclei segmentation and classification under limited annotations, data scarcity, and class imbalance. (MICCAI 2023, MICCAI 2024, ECCV 2024)
 - Developed weakly supervised and prompt-efficient segmentation methods with contrastive regularization and foundation-model analysis. (ISBI 2022, CIBM 2025, MedAGI 2023)
 - Designed diffusion-guided 3D CT super-resolution using 2D projection priors and Signed 3D Gaussians. (MICCAI 2026 Early Accept, top ~9%)
 - Developed diffusion augmentation method for data-scarce fine-grained classification using region-selective value attention and repulsive noise optimization.
 - Served as TA for Data Structures and Digital Image Processing; managed GPU servers; lab leader.
- Medical Information Processing Laboratory, Korea University** Mar. 2019 – Jun. 2019
Undergraduate Researcher Seoul, South Korea
 - Developed a virtual image reconstruction simulator for educational demonstrations in biomedical imaging.

EDUCATION

- Korea University** Mar. 2021 – Aug. 2026
Ph.D. in Computer Science Seoul, South Korea
 - GPA: 4.31/4.50 | Advisor: Won-Ki Jeong
 - Dissertation: *Controllable Diffusion-based Data Augmentation for Learning in Limited Data Regimes*
 - Ph.D. dissertation completed; degree expected in Aug. 2026.
- Korea University** Mar. 2014 – Feb. 2021
B.S. in Biomedical Engineering Seoul, South Korea
 - GPA: 4.03/4.50
- Chapman University** Spring 2020
Exchange Student (Computer Science & Biostatistics) Orange, CA, USA
- East China Normal University** Summer 2016
KU-China Global Leadership Program Shanghai, China
 - HSK Level 4 (264/300, Dec. 2021)

MILITARY SERVICE

- Republic of Korea Police** Oct. 2016 – Jul. 2018
Seoul Metropolitan Police Agency Seoul, South Korea

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, W=WORKSHOP

- [C.6] Jeonghyun Noh*, **Hyun-Jic Oh***, Won-Ki Jeong (2026). **Zero-shot CT Super-Resolution using Diffusion-based 2D Projection Priors and Signed 3D Gaussians**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2026)*, *Early Accept (top ~9%)*.
- [C.5] **Hyun-Jic Oh**, Junsik Kim, Zhiyi Shi, Yichen Wu, Yu-An Chen, Peter K. Sorger, Hanspeter Pfister, Won-Ki Jeong (2026). **Virtual Multiplex Staining for Histological Images using a Marker-wise Conditioned Diffusion Model**. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.
- [C.4] **Hyun-Jic Oh**, Won-Ki Jeong (2024). **Controllable and Efficient Multi-Class Pathology Nuclei Data Augmentation using Text-Conditioned Diffusion Models**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2024)*.
- [C.3] Seonghui Min*, **Hyun-Jic Oh***, Won-Ki Jeong (2024). **Co-synthesis of Histopathology Nuclei Image-Label Pairs using a Context-Conditioned Joint Diffusion Model**. In *European Conference on Computer Vision (ECCV 2024)*.
- [C.2] **Hyun-Jic Oh**, Won-Ki Jeong (2023). **DiffMix: Diffusion Model-based Data Synthesis for Nuclei Segmentation and Classification in Imbalanced Pathology Image Datasets**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)*.
- [C.1] **Hyun-Jic Oh**, Kanggeun Lee, Won-Ki Jeong (2022). **Scribble-Supervised Cell Segmentation Using Multiscale Contrastive Regularization**. In *IEEE International Symposium on Biomedical Imaging (ISBI 2022)*.
- [J.2] **Hyun-Jic Oh***, Seonghui Min*, Won-Ki Jeong (2025). **S2L-CM: Scribble-supervised Nuclei Segmentation in Histopathology Images using Contrastive Regularization and Pixel-Level Multiple Instance Learning**. *Computers in Biology and Medicine*, 2025.
- [J.1] Junyoung Choi, **Hyun-Jic Oh**, Hakjun Lee, Suyeon Kim, Seok-Kyu Kwon, Won-Ki Jeong (2023). **MitoVis: A Unified Visual Analytics System for End-to-End Neuronal Mitochondria Analysis**. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2023.
- [P.2] **Hyun-Jic Oh**, Won-Ki Jeong (2026). **Virtual Multiplex Staining with Marker-Aware Decoding and Test-Time LoRA Scaling**. *Preprint*.
- [P.1] **Hyun-Jic Oh***, Gayeon Koh*, Won-Ki Jeong (2026). **Improving Diffusion-Based Augmentation for Data-Scarce Image Classification via Mask-Guided Reference Attention and Repulsive Noise Optimization**. *Preprint*.
- [W.2] Gayeon Koh*, **Hyun-Jic Oh***, Jeonghyun Noh, Won-Ki Jeong (2025). **Synthetic Data Augmentation using Pre-trained Diffusion Models for Long-tailed Food Image Classification**. In *2nd MetaFood Workshop at CVPR 2025*.
- [W.1] Seungkyu Kim, **Hyun-Jic Oh**, Seonghui Min, Won-Ki Jeong (2023). **Evaluation and Improvement of Segment Anything Model for Interactive Histopathology Image Segmentation**. In *MICCAI 2023 1st International Workshop on Foundation Models for General Medical AI (MedAGI)*.

SKILLS

- **Programming Languages:** Python, MATLAB, C++
- **ML/AI Frameworks:** PyTorch, TensorFlow, MONAI, Hugging Face Diffusers
- **Imaging & Medical Data:** Histopathology, CT/MR imaging, multiplexed immunofluorescence, spatial omics, cellular/neuronal imaging
- **Tools & Platforms:** Linux, CUDA, Git, Docker, Slurm/HPC clusters, Weights&Biases, Hydra
- **Methods:** Diffusion models, generative modeling, data-centric augmentation, weakly supervised learning, contrastive learning, vision-language models, foundation models (SAM)

ACADEMIC SERVICES

- **Conference Reviewer**
 - CVPR, ECCV, NeurIPs, AAAI (2026)
 - MICCAI (2023–2026)
 - MICCAI Workshop (MedAGI) (2023-2024, 2026)
- **Journal Reviewer**
 - TMI, TMM, MIA, PR, EAAI, ESWA
- **Program Committee**
 - Student Contributor, PAIP 2023 AI Challenge: TC prediction in pancreatic and colon cancer, IEEE ISBI 2023

ADDITIONAL INFORMATION

Languages: English (Fluent), Korean (Native), Chinese (HSK Level 4)

Research Interests: Diffusion-based generative models, multimodal medical AI, synthetic data generation and validation, data-efficient visual learning, weakly supervised segmentation, foundation models

REFERENCES

1. **Prof. Won-Ki Jeong**

Associate Professor, Department of Computer Science and Engineering

Korea University, Seoul, South Korea

Email: wkjeong@korea.ac.kr

Relationship: Ph.D. advisor

2. **Prof. Hanspeter Pfister**

An Wang Professor of Computer Science

Harvard John A. Paulson School of Engineering and Applied Sciences

Email: pfister@seas.harvard.edu

Relationship: Host during visiting scholar period at Harvard VCG

3. **Dr. Yichen Wu**

Research Fellow, Department of Systems Biology

Harvard Medical School / Massachusetts General Hospital

Email: yichen_wu@hms.harvard.edu

Relationship: Collaborator on virtual multiplex staining project